Name: Rishi Agrawal **Roll No.:**C1-17

Practical No. 8

Aim:

Develop a Movie Review Sentiment Analysis system that classifies reviews as Positive or Negative using machine learning techniques. The system should include model comparison, GUI deployment, and GitHub project submission.

GitHub link: Movie_Review_Sentiment_Analysis-NLP

Dataset Used: Kaggle Link

Code:

```
import pandas as pd
import numpy as np
import re
import pickle
import matplotlib.pyplot as plt

import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.ensemble import RandomForestClassifier,
GradientBoostingClassifier
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score,
                             confusion_matrix, ConfusionMatrixDisplay)

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
df = pd.read_csv("IMDB Dataset.csv", engine='python',
                 on_bad_lines='skip', encoding='utf8')

print("Shape:", df.shape)
print("\nClass distribution:")
print(df['sentiment'].value_counts())
df.head()
```

```
stop_words  = set(stopwords.words('english'))
lemmatizer  = WordNetLemmatizer()

def preprocess(text):
    text   = re.sub(r'<.*?>', '', text)
    text   = text.lower()
    tokens = word_tokenize(text)
    tokens = [w for w in tokens if w.isalpha()]
    tokens = [w for w in tokens if w not in stop_words]
    tokens = [lemmatizer.lemmatize(w) for w in tokens]
    return ' '.join(tokens)

df['clean_text'] = df['review'].apply(preprocess)
df['sentiment']  = df['sentiment'].map({'positive': 1, 'negative': 0})

print("Preprocessing done!")
print(df[['review', 'clean_text']].head(3))
```

```
vectorizer = TfidfVectorizer(
    max_features  = 50000,
    ngram_range   = (1, 2),
    sublinear_tf  = True,
    min_df        = 2,
    strip_accents = 'unicode'
)

X = vectorizer.fit_transform(df['clean_text'])
```

```
y = df['sentiment']

print("Feature matrix shape:", X.shape)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Train: {X_train.shape[0]} | Test: {X_test.shape[0]}")
```

```
models = {
    "Naive Bayes"         : MultinomialNB(alpha=0.1),
    "Logistic Regression" : LogisticRegression(C=10, max_iter=1000,
solver='lbfgs', n_jobs=-1),
    "SVM"                 : LinearSVC(C=1.0, max_iter=2000),
    "Random Forest"       : RandomForestClassifier(n_estimators=200,
random_state=42, n_jobs=-1),
    "Gradient Boosting"   : GradientBoostingClassifier(n_estimators=100,
random_state=42)
}

results = []

for name, model in models.items():
    print(f"Training {name}...", end=" ")
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    results.append({
        "Model"     : name,
        "Accuracy"  : round(accuracy_score(y_test, y_pred),  4),
        "Precision" : round(precision_score(y_test, y_pred), 4),
        "Recall"    : round(recall_score(y_test, y_pred),    4),
        "F1 Score"  : round(f1_score(y_test, y_pred),        4)
    })
    print(f"✅ Accuracy: {results[-1]['Accuracy']*100:.2f}%")
```

```
result_df = pd.DataFrame(results).sort_values('Accuracy',
ascending=False).reset_index(drop=True)
result_df
```

```
x     = range(len(result_df))
width = 0.2

fig, ax = plt.subplots(figsize=(13, 5))
ax.bar([i-0.3 for i in x], result_df['Accuracy'],  width,
label='Accuracy',  color='steelblue')
ax.bar([i-0.1 for i in x], result_df['Precision'], width,
label='Precision', color='tomato')
ax.bar([i+0.1 for i in x], result_df['Recall'],    width, label='Recall',
color='seagreen')
ax.bar([i+0.3 for i in x], result_df['F1 Score'],  width, label='F1
Score',  color='orange')

ax.set_xticks(x)
ax.set_xticklabels(result_df['Model'], rotation=15, fontsize=11)
ax.set_ylim(0.7, 1.02)
ax.set_title('Model Comparison — All Metrics', fontsize=14,
fontweight='bold')
ax.legend()
plt.tight_layout()
plt.show()
```

```
best_row   = result_df.iloc[0]
best_name  = best_row['Model']
best_model = models[best_name]

print(f"Best Model  : {best_name}")
print(f"Accuracy    : {best_row['Accuracy']*100:.2f}%")
print(f"Precision   : {best_row['Precision']}")
print(f"Recall      : {best_row['Recall']}")
print(f"F1 Score    : {best_row['F1 Score']}")

y_pred_best = best_model.predict(X_test)
```

```
cm   = confusion_matrix(y_test, y_pred_best)
disp = ConfusionMatrixDisplay(confusion_matrix=cm,
                              display_labels=['Negative', 'Positive'])
disp.plot(cmap='Blues')
plt.title(f'Confusion Matrix — {best_name}')
plt.show()
```

```
with open('best_model.pkl', 'wb') as f:
    pickle.dump(best_model, f)

with open('tfidf_vectorizer.pkl', 'wb') as f:
    pickle.dump(vectorizer, f)

print(f"Saved best_model.pkl  ({best_name})")
print("Saved tfidf_vectorizer.pkl")
```

```
!pip install gradio -q

import gradio as gr
import pickle

with open('best_model.pkl', 'rb') as f:
    gui_model = pickle.load(f)

with open('tfidf_vectorizer.pkl', 'rb') as f:
    gui_vec = pickle.load(f)

def predict_sentiment(review):
    cleaned    = preprocess(review)
    features   = gui_vec.transform([cleaned])
    prediction = gui_model.predict(features)[0]
    return "🟢 Positive" if prediction == 1 else "🔴 Negative"

demo = gr.Interface(
    fn          = predict_sentiment,
    inputs      = gr.Textbox(lines=5, placeholder="Paste a movie review
here..."),
```

```
    outputs     = gr.Textbox(label="Predicted Sentiment"),
    title       = "🎬 Movie Review Sentiment Analyser",
    description = f"Best Model: {best_name} | Trained on IMDB 50K Dataset"
)

demo.launch(share=True)
```

Output:

```
df = pd.read_csv('IMDB Dataset.csv', engine='python',
                 on_bad_lines='skip', encoding='utf8')

print("Shape:", df.shape)
print("\nClass distribution:")
print(df['sentiment'].value_counts())
df.head()
```

```
Shape: (50000, 2)

Class distribution:
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

Next steps: Generate code with df New interactive sheet

Name: Rishi Agrawal **Roll No.:**C1-17

```python
stop_words  = set(stopwords.words('english'))
lemmatizer  = WordNetLemmatizer()

def preprocess(text):
    text   = re.sub(r'<.*?>', '', text)
    text   = text.lower()
    tokens = word_tokenize(text)
    tokens = [w for w in tokens if w.isalpha()]
    tokens = [w for w in tokens if w not in stop_words]
    tokens = [lemmatizer.lemmatize(w) for w in tokens]
    return ' '.join(tokens)

df['clean_text'] = df['review'].apply(preprocess)
df['sentiment']  = df['sentiment'].map({'positive': 1, 'negative': 0})

print(df[['review', 'clean_text']].head(3))
```

```
                                                 review  \
0  One of the other reviewers has mentioned that ...
1  A wonderful little production. <br /><br />The...
2  I thought this was a wonderful way to spend ti...

                                            clean_text
0  one reviewer mentioned watching oz episode hoo...
1  wonderful little production filming technique ...
2  thought wonderful way spend time hot summer we...
```

```python
vectorizer = TfidfVectorizer(
    max_features  = 50000,
    ngram_range   = (1, 2),
    sublinear_tf  = True,
    min_df        = 2,
    strip_accents = 'unicode'
)

X = vectorizer.fit_transform(df['clean_text'])
y = df['sentiment']

print("Feature matrix shape:", X.shape)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Train: {X_train.shape[0]} | Test: {X_test.shape[0]}")
```

```
Feature matrix shape: (50000, 50000)
Train: 40000 | Test: 10000
```

Name: Rishi Agrawal **Roll No.:**C1-17

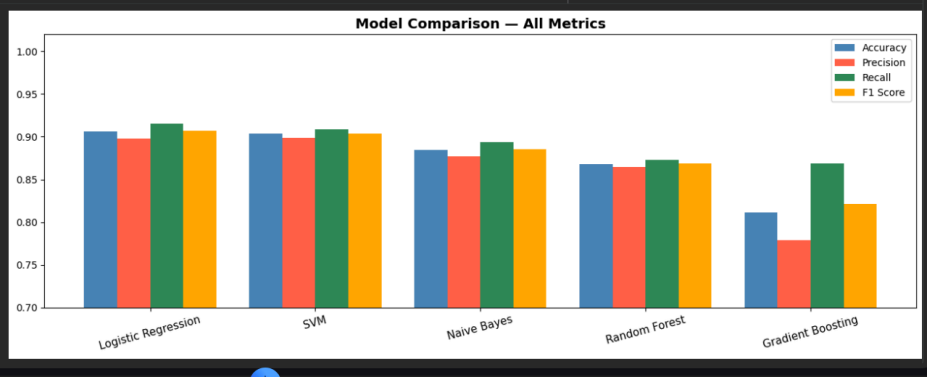
```
[5]
6m
            "Precision" : round(precision_score(y_test, y_pred), 4),
            "Recall"    : round(recall_score(y_test, y_pred),    4),
            "F1 Score"  : round(f1_score(y_test, y_pred),        4)
        })
        print(f"✅ Accuracy: {results[-1]['Accuracy']*100:.2f}%")

    result_df = pd.DataFrame(results).sort_values('Accuracy', ascending=False).reset_index(drop=True)
    result_df
```

```
Training Naive Bayes... ✅ Accuracy: 88.45%
Training Logistic Regression... ✅ Accuracy: 90.59%
Training SVM... ✅ Accuracy: 90.35%
Training Random Forest... ✅ Accuracy: 86.78%
Training Gradient Boosting... ✅ Accuracy: 81.09%
```

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.9059	0.8982	0.9156	0.9068
1	SVM	0.9035	0.8991	0.9090	0.9040
2	Naive Bayes	0.8845	0.8773	0.8940	0.8856
3	Random Forest	0.8678	0.8643	0.8726	0.8684
4	Gradient Boosting	0.8109	0.7788	0.8684	0.8212

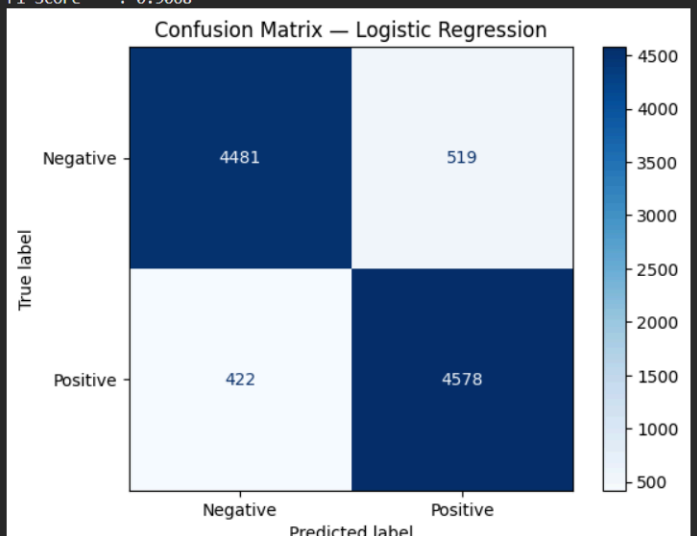
Next steps: Generate code with result_df | New interactive sheet

```
ax.set_xticklabels(result_df['Model'], rotation=15, fontsize=11)
ax.set_ylim(0.7, 1.02)
ax.set_title('Model Comparison — All Metrics', fontsize=14, fontweight='bold')
ax.legend()
plt.tight_layout()
plt.show()
```

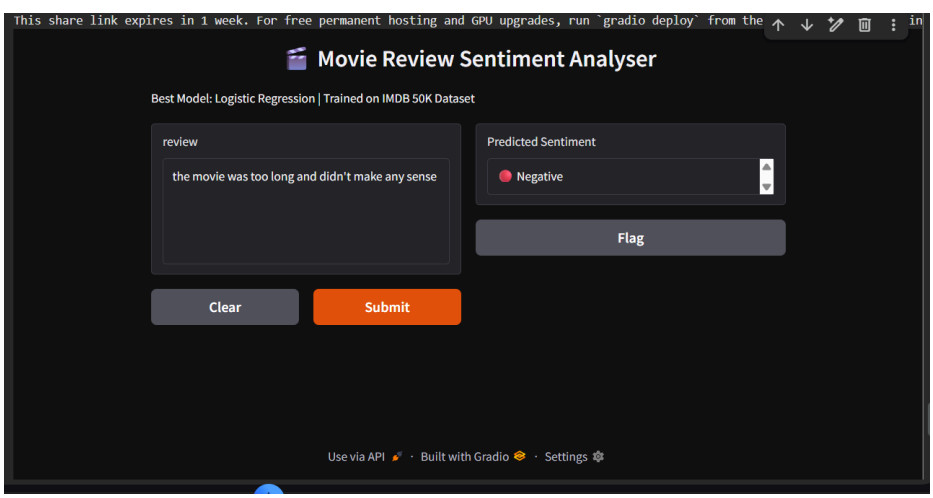


Name: Rishi Agrawal **Roll No.:**C1-17

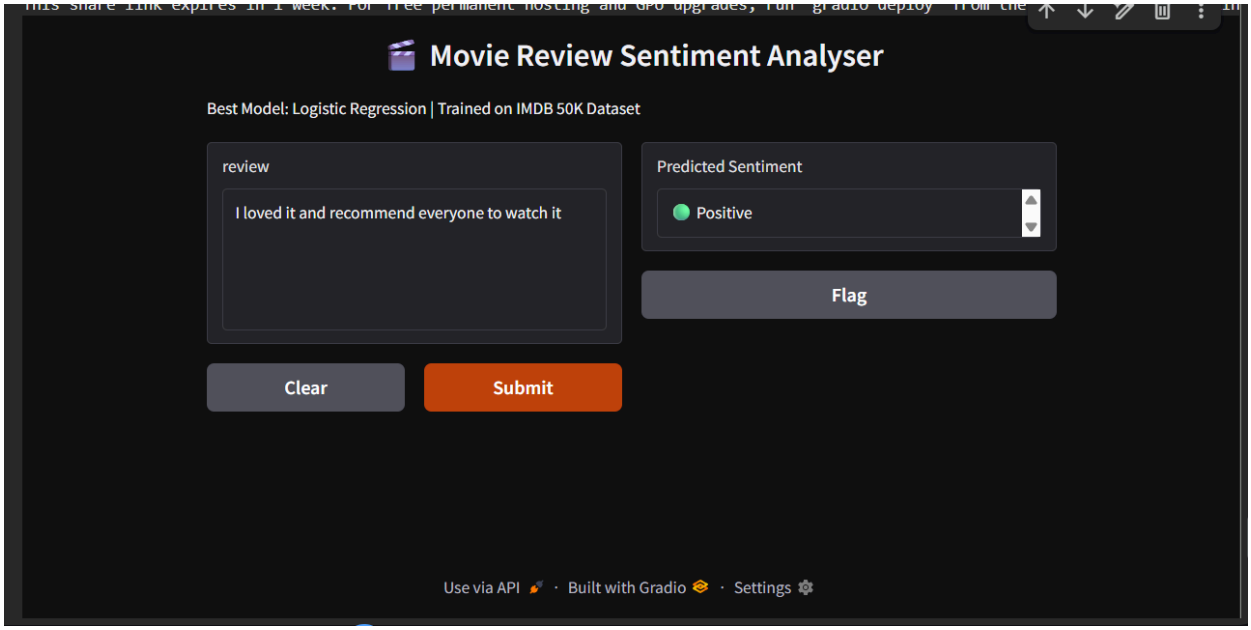
```
Best Model   : Logistic Regression
Accuracy     : 90.59%
Precision    : 0.8982
Recall       : 0.9156
F1 Score     : 0.9068
```




```
[8]
✓ 0s
with open('best_model.pkl', 'wb') as f:
    pickle.dump(best_model, f)

with open('tfidf_vectorizer.pkl', 'wb') as f:
    pickle.dump(vectorizer, f)

print(f"Saved best_model.pkl  ({best_name})")
print("Saved tfidf_vectorizer.pkl")

Saved best_model.pkl  (Logistic Regression)
Saved tfidf_vectorizer.pkl
```



Name: Rishi Agrawal **Roll No.:**C1-17



https://34752a4a6b4c3de835.gradio.live

Conclusion:

Thus we have successfully learned how use GUIs and also created a basic movie review sentiment analysis tool.